

Ultimate Nutrition

Coenzyme Q-10

Energy is the currency the body uses to carry out all metabolic processes and to function optimally. Energy is especially required during exercise and in endurance training when the athletes not only need instant energy but also a sufficient supply of oxygen to maintain a rigorous regimen. Coenzyme Q10 is the spark that produces energy in the body.

Coenzyme Q exists in every cell in the body. It functions as a messenger in the units called mitochondria, which are the powerhouses of the cell, where the oxidation of the foods eaten takes place and energy is generated as a result. This process is usually called respiration. Even though the body can synthesize CoQ10, it can be deficient in CoQ10 in cases of increased tissue needs, such as during high intensity exercise regimens. If respiration is limited, the body does not produce sufficient energy. Therefore, the blood supply to various tissues and organs is impaired. Nutritional supplementation with CoQ10 can replenish it under conditions of extreme duress, such as in endurance training.

Because CoQ10 produces energy in combination with respiration, supplementation with dietary CoQ10 ensures that muscles and other critical tissues and organs receive ample amount of oxygen via blood. Since the blood also carries nutrients in its stream, delivery of oxygen also provides appropriate nutrition to muscles and other organs critical for the bodily metabolism and survival.

In addition to its critical role in energy production "on demand," CoQ10 is one of the more potent natural antioxidant found in nature. As such, it protects proteins, membranes and nucleic acids (DNA and RNA, the molecules that make up the genetic material) from the ravages of free radicals.

The importance of CoQ10 can hardly be overemphasized. In its role as the agent that produces oxygen, CoQ10 is extremely important for heart health. In fact, deficiency of CoQ10 has been shown to be a causative factor in a variety of illnesses.

Ultimate Nutrition's 100% Premium Coenzyme Q10 is designed and manufactured for athletes who have high intensity exercise regimen. To supply the muscle and other tissues with the right amount of oxygen and nutrients and to produce energy when it is needed, CoQ10 is the supplemental nutrient of choice.

SELECTED REFERENCES

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FREQUENTLY ASKED QUESTIONS

1. What is a Coenzyme?

The key functions of nutrients such as vitamins and minerals in the human body is to provide support to enzymes and coenzymes. Enzymes are molecules that are involved in speeding up the chemical reactions necessary for human physiology. As such, enzymes function as catalysts. Coenzymes, on the other hand, are molecules that help the enzymes in their function as catalysts.

Enzymes and coenzymes work to either join molecules together or split them apart by breaking the chemical bonds between them. Thus, CoQ10 supports numerous enzymes in the mitochondria to break down food and generate energy.

2. What is CoQ10?

The human body consists of more than a trillion cells. Each of these cells has mitochondria, which produce energy. It is in mitochondria that the nutrients obtained from the foods are burned in the presence of the oxygen you breathe. In order to produce energy, mitochondria must have CoQ10, which assists several enzymes in this process.

3. If CoQ10 "Burns" Food, Does it Help in Reducing Weight?

Yes, indeed! Coenzyme Q10 is an essential compound necessary for the transport and breakdown of fat into energy. Clinical studies have corroborated this contention. Thus, in one clinical trial, supplementation with CoQ10 in overweight subjects led to 29.7 pounds of weight loss on the average. This study suggests that overweight people may be deficient in CoQ10 and dietary supplementation with 100 mg daily may help lose weight along with a reasonable exercise regimen.

4. What is the Daily Dose of CoQ10?

Numerous studies have shown that, as a nutritional supplement, a 30mg CoQ10 dosage

is sufficient for people to observe beneficial effects. The dosage can be increased in consultation with the primary care physician.

5. Why Does CoQ10 Have Antioxidant Properties?

During the energy production, free radicals are produced as a matter of course. Therefore, it is understandable that the nutrient bringing the fats into the mitochondria should be equipped with the ability to scavenge free radicals.